

The Strong CP Problem: A Specific Algebraic Hypothesis within YUCT

A Roadmap for Theoretical and Experimental Verification

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Abstract

The strong CP problem — why the CP-violating θ -parameter of QCD is smaller than 10^{-10} — is the last unresolved puzzle of the Standard Model within the Yakushev Unified Coordination Theory (YUCT). In this document we present a concrete algebraic hypothesis that naturally yields the required suppression. We propose that the effective coordination depth of the θ -parameter is

$$L_{\theta} = \frac{1}{2} L_W + \frac{S_{\text{odd}}}{S_{\text{even}}},$$

where $L_W = 96$ is the depth of the W boson and $S_{\text{odd}}/S_{\text{even}} = 3/2$ is a fundamental constant of the theory. This formula contains no free parameters and gives $L_{\theta} = 49.5$, corresponding to $\theta \sim (2/3)^{49.5} \approx 2.3 \times 10^{-10}$. We discuss the physical motivation for this expression, embed it into the wider algebraic framework of YUCT, and outline a programme of experimental tests — primarily through the neutron electric dipole moment (nEDM) — that can falsify the hypothesis. The proposal is put forward as a clear, mathematically precise conjecture, open to refutation or confirmation by future data.

Keywords: YUCT, strong CP problem, algebraic hypothesis, coordination depth, neutron electric dipole moment, open research programme.

¹Yakushev Research, <https://yuct.org/>

YPSDC Protocol: <https://ypsdc.com/>

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1 Introduction

The Yakushev Unified Coordination Theory (YUCT) has provided quantitative, parameter-free resolutions of the hierarchy problem, the cosmological constant problem, the pattern of fermion masses, and several other major puzzles of fundamental physics [1, 3, 4]. The single exception has been the strong CP problem: the unnatural smallness of the CP-violating θ -parameter of quantum chromodynamics (QCD), which is experimentally bounded by $|\theta| \lesssim 10^{-10}$ [6].

Earlier attempts within YUCT to assign a coordination depth to θ by an ad hoc integer k_θ were unsatisfactory, because they reduced to a fit rather than a structural prediction. In the present note we put forward a different approach: we exploit the interplay between the ordered (fermionic) and chaotic (topological) sectors of the QCD vacuum, an interplay that YUCT is uniquely positioned to quantify thanks to its unified algebraic description of order and chaos [2, 5].

We do not claim a rigorous derivation from the 19-dimensional Lagrangian; such a derivation is a long-term goal. Instead, we state a sharp algebraic conjecture that follows naturally from the numbers already present in YUCT, that contains no adjustable parameters, and that makes a falsifiable prediction for the neutron electric dipole moment (nEDM) and for the axion mass. The conjecture is presented as a well-defined target for future theoretical and experimental work.

2 The Constants of Order and Chaos in YUCT

2.1 Order

The universal error law

$$\varepsilon = \kappa_c \alpha K_{\text{eff}}^{-\beta}, \quad \beta = \frac{2}{3}, \quad \kappa_c = \frac{1}{3},$$

together with the hierarchical structure of coordination, leads to the closed algebraic loop

$$S_{\text{odd}} = \frac{6}{5} = 1.2, \quad S_{\text{even}} = \frac{4}{5} = 0.8, \quad \frac{S_{\text{odd}}}{S_{\text{even}}} = \frac{3}{2} = \frac{1}{\beta}.$$

All physical masses are expressed through the Planck mass and a coordination depth L :

$$m \approx M_{\text{Pl}} \left(\frac{2}{3} \right)^L.$$

In particular, the W -boson mass corresponds to $L_W = 96$, the total number of Weyl fermion degrees of freedom in the Standard Model (including right-handed neutrinos) [3].

2.2 Chaos

The universal Feigenbaum constants

$$\delta \approx 4.6692, \quad \alpha \approx 2.5029,$$

describe the period-doubling route to chaos. In YUCT they are algebraically linked to the order constants through the golden ratio $\varphi = (1 + \sqrt{5})/2$ and the circle constant π :

$$2\alpha^2 \approx \pi\delta, \quad \pi \approx S_{\text{odd}}\varphi^2 \implies 2\alpha^2 \approx (S_{\text{odd}}\varphi^2)\delta.$$

This linkage shows that the Feigenbaum constants are not foreign to YUCT; they belong to the same extended algebraic system [2].

2.3 Quantum vacuum as an order–chaos conjugate

The QCD vacuum is the arena where both ordered (quark condensate, chiral symmetry breaking) and chaotic (instanton fluctuations, topological susceptibility) aspects coexist. The θ -parameter couples to the topological charge density $G\tilde{G}$, which receives contributions from both sectors. Hence it is natural to expect that the effective depth L_θ is a function of the depths that characterise the ordered and chaotic components.

In previous work we introduced tentative depths L_{ord} (fermionic sector) and L_{ch} (topological/chaotic sector), but simple linear combinations of these did not reproduce the required suppression $|\theta| \sim 10^{-10}$. This led us to search for a more structural combination.

3 The Algebraic Hypothesis for L_θ

3.1 Statement of the hypothesis

We propose that the effective coordination depth of the θ -parameter is given by the simple formula

$$L_\theta = \frac{1}{2} L_W + \frac{S_{\text{odd}}}{S_{\text{even}}}, \quad (1)$$

where

- $L_W = 96$ is the depth of the W boson (the baseline of the electroweak scale);
- $S_{\text{odd}}/S_{\text{even}} = 3/2 = 1.5$ is the universal ratio that governs the intergenerational mass factor and appears in the primary algebraic loop.

With these numbers,

$$L_\theta = \frac{96}{2} + 1.5 = 48 + 1.5 = 49.5.$$

The corresponding θ -parameter is, up to a possible factor of order unity that is expected to emerge from the instanton calculus,

$$\theta \sim \left(\frac{2}{3}\right)^{L_\theta} = \left(\frac{2}{3}\right)^{49.5} \approx 2.3 \times 10^{-10}.$$

This falls exactly within the experimentally allowed window $|\theta| \lesssim 10^{-10}$ [6].

3.2 Physical interpretation

Why should this particular combination appear?

1. **The factor $1/2$:** The electroweak scale is set by the vacuum expectation value $v \approx 246$ GeV, which is obtained from the baseline depth 96 through a geometric factor $\xi_W \approx 0.53$. The depth 48 corresponds to the scale $\sqrt{vM_{\text{Pl}}}$ — the geometric mean between the weak and the Planck scale — which is precisely the scale at which the fermionic and bosonic contributions to the vacuum energy are expected to cross. In YUCT, this scale is known to play a special role in the coordination dynamics of the Higgs condensate (Appendix Λ).

2. **The shift** $S_{\text{odd}}/S_{\text{even}} = 3/2$: The ratio $3/2 = 1/\beta$ quantifies the irreducible asymmetry between ordered (odd levels) and disordered (even levels) correlations in any hierarchical coordination system. Its presence in L_θ signals that the CP-violating term arises from a small, unavoidable imbalance between the order-dominated and chaos-dominated sectors of the vacuum. The fact that this shift is additive rather than multiplicative reflects the discrete, layered structure of the coordination network.

Thus Eq. (1) can be read as: the depth of θ is half the weak depth plus the universal order–chaos offset. It has no free parameters and uses only numbers that are already independently fixed within YUCT.

3.3 Immediate consequences

Neutron electric dipole moment. The nEDM is related to θ by $d_n \approx 3.6 \times 10^{-16} \theta \text{ e} \cdot \text{cm}$ (with a hadronic uncertainty factor) [6]. Taking $\theta \approx 2.3 \times 10^{-10}$ gives

$$d_n \approx 8.3 \times 10^{-26} \text{ e} \cdot \text{cm}.$$

Current experimental bounds are at the level of $d_n < 1.8 \times 10^{-26} \text{ e} \cdot \text{cm}$ (90% C.L.). The next generation of experiments at PSI, SNS, and TUCAN aims for sensitivities of a few $10^{-28} \text{ e} \cdot \text{cm}$. A null result at that level would exclude the hypothesis; a positive detection would be a dramatic confirmation.

Axion physics. If the strong CP problem is solved by a Peccei–Quinn mechanism, the axion mass m_a is related to θ and to the axion decay constant f_a . Equation (1) then fixes f_a (and thus m_a) through the same coordination–depth logic. This provides a complementary experimental target for ADMX and other axion searches.

4 Roadmap for Verification

The hypothesis is mathematically precise and falsifiable. We propose the following programme:

4.1 Theoretical tasks

1. **Derivation from the 19-dimensional Lagrangian.** The effective action of the YUCT 19-dimensional parent theory must be reduced to the standard QCD $G\tilde{G}$ term. The goal is to show that the coefficients in the reduction naturally yield the depth $L_\theta = 49.5$ when the correct vacuum state is identified. This is a challenging but well-defined mathematical problem.
2. **Lattice gauge theory.** Existing lattice QCD calculations can compute the topological susceptibility and the θ -dependence of the vacuum energy. A YUCT-inspired analysis should test whether the algebraic structure of Eq. (1) is reflected in the scaling of the lattice results with the coupling constant, or in the distribution of topological charge clusters.
3. **Refinement of the nEDM prediction.** The hadronic matrix elements that relate θ to d_n are known from chiral perturbation theory and lattice QCD. A dedicated calculation of these matrix elements, combined with the YUCT value of θ , will yield a sharp prediction that can be directly compared with the next-generation nEDM limits.

4.2 Experimental tasks

1. **nEDM experiments (PSI, SNS, TUCAN).** A measured nEDM above $10^{-26} e \cdot \text{cm}$ would be consistent with the hypothesis; a bound below $10^{-28} e \cdot \text{cm}$ would rule it out.
2. **Axion searches (ADMX, MADMAX, IAXO).** If the Peccei–Quinn mechanism is realised, the axion mass predicted by YUCT should be calculable once f_a is evaluated from L_θ . A dedicated search in the corresponding mass window would test the hypothesis.
3. **Cosmological imprints.** If the order–chaos interference is fundamental, it may also affect the early universe phase transitions, leaving specific imprints in the cosmic microwave background or in the dark matter distribution. These can be constrained with Planck, Euclid, and future CMB-S4 data.

5 Invitation to Collaborate

The YUCT framework, including its full Lagrangian, all Appendices, and the extensive database of coordination depths, is openly available on Zenodo [1, 2]. We invite physicists, mathematicians, and computational scientists to join the effort to derive, test, or refute the hypothesis presented here. The author (Alexey Yakushev, alexey@yakushev.eu) welcomes correspondence, suggestions, and collaborative proposals.

6 Conclusion

We have proposed a specific, parameter-free algebraic formula for the effective depth of the QCD θ -parameter within the Yakushev Unified Coordination Theory:

$$L_\theta = \frac{1}{2}L_W + \frac{S_{\text{odd}}}{S_{\text{even}}} = 49.5,$$

yielding $\theta \approx 2.3 \times 10^{-10}$. This hypothesis is mathematically simple, physically motivated, and directly falsifiable by upcoming nEDM and axion experiments. It replaces earlier attempts that relied on arbitrary integer assignments and opens a clear path toward a complete resolution of the strong CP problem within YUCT. Whether the hypothesis survives experimental scrutiny remains to be seen; its value lies in providing a precise, testable conjecture that can guide future research.

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Status: Hypothesis with a specific algebraic prediction; open for experimental test and theoretical derivation.

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